SCC.160: Fundamentals of Communication Systems Tutorial 2

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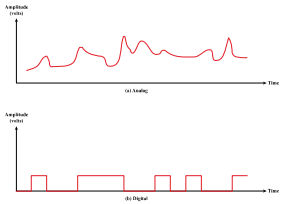
Question 1

Differentiate between an analogue and a digital electromagnetic signal.

A continuous or analog signal is one in which the signal intensity varies in a smooth fashion over time while a discrete or digital signal is one in which the signal intensity maintains one of a finite number of constant levels for some period of time and then changes to another constant level.



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Question 2

What are three important characteristics of a periodic signal?

A periodic signalcan be represented by three parameters: ***peak amplitude, frequency, phase***

$y(t) = A * \sin(2\pi f\, t + \phi)$

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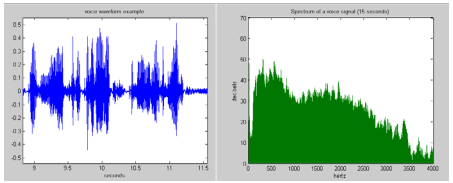
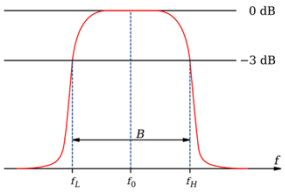
Question 3

What is the relationship between a signal's spectrum and its bandwidth?

The spectrum of a signal consists of the frequencies it contains; the bandwidth of a signal is the width of the spectrum (defined by the frequency that is half-attenuated).





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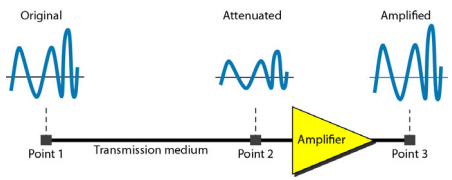
Question 4

What is attenuation?

Attenuation is the gradual weakening of a signal over distance.



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Question 5

Define channel capacity.

The maximum rate at which information can be transmitted over a given communication path, or channel, under given conditions, is referred to as the channel capacity.

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引导介质为固体
非引导介质为液体，气体

Question 6

What key factors affect channel capacity?

Bandwidth, noise, and error rate affect channel capacity.
The bandwidth of the channel refers to the range of frequencies that can be transmitted through the channel. A wider bandwidth means that more data can be transmitted, resulting in a higher channel capacity.

The signal-to-noise ratio (SNR) is a measure of the quality of the signal in the channel. It is defined as the ratio of the signal power to the noise power. A higher SNR means that the signal is stronger than the noise, resulting in a higher channel capacity.

Error rates refer to the number of errors that occur during transmission. These errors can be caused by a variety of factors, including interference, noise, and distortion. A lower error rate means that more data can be transmitted without errors, resulting in a higher channel capacity.

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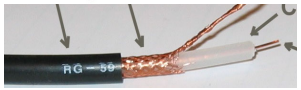
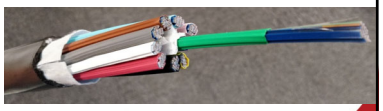
Question 7

Differentiate between guided media and unguided media.

With guided media, the electromagnetic waves are guided along an enclosed physical path, whereas unguided media provide a means for transmitting electromagnetic waves through space, air, or water, but do not guide them.





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Question 8

What are some major advantages and disadvantages of microwave transmission?

Point-to-point microwave transmission has a high data rate and less attenuation than twisted pair or coaxial cable.

It is affected by rainfall, however, especially above 10 GHz.

It is also requires line of sight and is subject to interference from other microwave transmission, which can be intense in some places.

衰减 $= 10 \times \lg \frac{P_2}{P_1}$

$= 10 \times \lg \frac{0.333}{1}$

$= 10 \times \lg 0.333$

$= 10 \times (-0.4775)$

$= -4.775\ dB$

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$$\text{Attenuation} = 10 \times \lg \frac{P_{out}}{P_{in}}$$

衰减是负数

Question 9

The power of a signal is 10 mW and the power of the noise is 1 μW. What are the values of SNR and SNR_{dB}?

Micro-watt, abbreviated as μW, is 10^-6 watt.

The values of SNR and SNR_{dB} can be calculated a

SNR = 10000 μW / 1 μW = 10000

SNR_{dB} = 10 $\log_{10}$ (10000) = 40 dB

SI multiples of watt (W)

Submultiples			Multiples		
Value	SI symbol	Name	Value	SI symbol	Name
10^{-1} W	dW	deciwatt	10^{1} W	daW	decawatt
10^{-2} W	cW	centiwatt	10^{2} W	hW	hectowatt
10^{-3} W	**mW**	**milliwatt**	10^{3} W	**kW**	**kilowatt**
10^{-6} W	**μW**	**microwatt**	10^{6} W	**MW**	**megawatt**
10^{-9} W	**nW**	**nanowatt**	10^{9} W	**GW**	**gigawatt**
10^{-12} W	**pW**	**picowatt**	10^{12} W	**TW**	**terawatt**
10^{-15} W	fW	femtowatt	10^{15} W	**PW**	**petawatt**
10^{-18} W	aW	attowatt	10^{18} W	EW	exawatt
10^{-21} W	zW	zeptowatt	10^{21} W	ZW	zettawatt
10^{-24} W	yW	yoctowatt	10^{24} W	YW	yottawatt

Common multiples are in **bold face**

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Question 10

*Suppose a signal travels through a transmission medium and its power is reduced to one-third. This means that P2 = 0.333*P1. Find the attenuation (loss of power) for this case (measured in dB).*

$10\log_{10}(P2/P1)$

$= 10 \log_{10}(0.333*P1/P1)$

$= 10 \log_{10}(0.333)$

$= 10*(-0.4775)$

$= -4.775$ dB

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Question 11

*A signal travels through an amplifier, and its power is increased 5 times. This means that P2 = 5*P1. Find the amplification (gain of power in dB) for this case.*

$10\log_{10}(P2/P1)$

$= 10 \log_{10}(5*P1/P1)$

$= 10 \log_{10}(5)$

$= 10*(0.6989)$

$= 6.989$ dB

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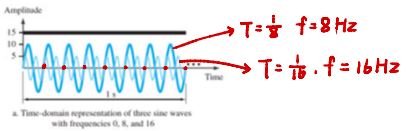
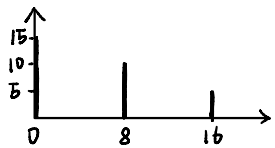
Question 12

The below figure shows three sine waves, each with different amplitude and frequency. Please represent these three sine waves in the frequency domain.





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注意单位

$SNRdB = 10 \cdot \lg(SNR)$

$3 = 10 \cdot \lg(SNR)$

$0.3 = \lg(SNR)$

$SNR = 10^{0.3}$

$C = 300 \cdot \log_2(1+10^{0.3})$

$= 474\text{bps}.$

Question 13

What is the channel capacity for a teleprinter channel with a 300-Hz bandwidth and a signal-to-noise ratio of 3 dB?

Using Shannon's equation:
$C = B\log_2(1 + SNR)$

We have
$B = 300$ Hz
$(SNR)_{dB} = 3$
Therefore, $SNR = 10^{0.3}$

$C = 300\log_2(1 + 10^{0.3})$
$= 300\log_2(2.995)$
$= 474$bps

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In order to transform these three Sine Waves from the time domain to the frequency domain, it is important to know two things which are

1) Frequency of each Sine Wave
2) Amplitude of each Sine Wave

From the above figure, we get that

- The first Sine Wave (the black horizontal line) does not change over time so its frequency is zero and amplitude is 15.
- The second Sine Wave repeat itself 8 times in the 1 second so its frequency is 8 Hz and amplitude is 10.
- The third Sine Wave repeat itself 16 times in the 1 second so its frequency is 16 Hz and amplitude is 5.

We can then plot the frequency in Hz on the x-axis and label the x-axis as Frequency [Hz] and plot the Amplitude in Volts on the y-axis and label the y-axis as Amplitude [V].

We will represent each Sine Wave as a spike with its corresponding frequency on x-axis and the amplitude on y-axis, as shown in the following figure. For example, for first sine wave, a spike will be plotted a 0 Hz with an amplitude of 15 volts.

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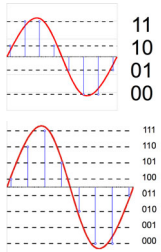
说明 $L = 2^4$
$\log_2 L = 4$

说明 $L = 2^8$
$\log_2 L = 8$

Question 14

A digital signalling system is required to operate at 9600 bps.

a) If a signal element encodes a 4-bit word, what is the minimum required bandwidth of the channel?
b) Repeat part (a) for the case of 8-bit words.



Using Nyquist's equation: $C = 2.\ B.\log_2 L$
We have $C = 9600$bps

a. $L = 4$, because a signal element encodes a 4 -bit word
Therefore, $C = 9600 = 2B \times 4$, and $B = 1200$ Hz
b. $9600 = 2\ B \times 8$, and $B = 600$ Hz

C

$N = 2B \cdot r$

$\boxed{S = 2B}$

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$20\times10^6 = 3\times10^6 \cdot \log_2(1+SNR)$

$\log_2(1+SN$

Question 15

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Study the works of Shannon and Nyquist on channel capacity. Each places an upper limit on the bit rate of a channel based on two different approaches. How are the two related?

Nyquist analysed the theoretical capacity of a noiseless channel; therefore, in that case, the Bit rate (Channel Capacity) is limited solely by channel bandwidth.

Bit Rate = 2 . B . $\log_2$ L

Shannon addressed the question of what Bit rate (Channel Capacity) can be achieved over a channel with a given bandwidth, a given signal power, and in the presence of

C = B . $\log_2$(1+SNR)

S C

$Gain = 35-12-10 = 13$

$13 = 10\cdot\lg\left(\frac{P_{out}}{4\times10^{-6}}\right)$

$P_{out} = 4\times10^{1.3}$ mW

$= 79.8$ mV

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Question 16

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Given a channel with an intended capacity of 20 Mbps, the bandwidth of the channel is 3 MHz. What signal-to-noise ratio is required to achieve this capacity?

$$C = B\log_2(1+SNR)$$

$$20\times10^6 = 3\times10^6\times\log_2(1+SNR)$$

$$\log 2(1+SNR) \Rightarrow 6.67$$

$$1+SNR \Rightarrow 102$$

$$SNR \Rightarrow 101$$

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Question 17

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Consider a series in which the input is at a power level of 4 mW, the first element is a transmission line with a 12-dB loss (-12 dB gain), the second element is an amplifier with a 35-dB gain, and the third element is a transmission line with a 10-dB loss.Calculate the net gain and output power.

$Gain_{\text{db}} = 13 = 10\log(P_{\text{out}}/4\text{ mW})$

$P_{\text{out}} = 4\times10^{1.3}\text{ mW} = 79.8\text{ mW}$

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Thank you

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